

BSc CSIT – 3rd Semester

30-Day Board Exam Study Plan

March 6 (Fagun 22) → April 4 (Chaitra 21)

■ All 5 Subjects • Every Day • Unit-wise Topics & Subtopics

Subject Color Guide

DSA	Data Structures & Algorithms
NM	Numerical Methods
CA	Computer Architecture
CG	Computer Graphics
STAT	Statistics II

■ Daily Study Schedule

Morning (2 hrs): DSA + Numerical Method | Afternoon (2 hrs): Computer Architecture + Computer Graphics | Evening (2 hrs): Statistics II + Revision

Use Pomodoro: 50 min study → 10 min break. Write notes, don't just read. Solve past questions!

Week 1 – Foundation (Days 1–7)

Day 1 | March 6 (Fagun 22)

DSA Data Structures & Algorithms	Unit 1 ■ Introduction to Data Structures Data types vs Data Structures, Abstract Data Type (ADT), Static vs Dynamic Memory, malloc/calloc/free in C
NM Numerical Methods	Unit 1 ■ Errors in Numerical Computation Sources of error, Round-off error, Truncation error, Absolute & Relative error, Significant digits
CA Computer Architecture	Unit 1 ■ Data Representation Binary representation, BCD, Complements (1's & 2's), Fixed-point & Floating-point, Overflow detection
CG Computer Graphics	Unit 1 ■ Introduction to Computer Graphics Areas of application (CAD, Gaming, Simulation, Medical), Display technology overview
STAT Statistics II	Unit 1 ■ Sampling Distribution Population vs Sample, Sampling distribution of mean, Standard error, Central Limit Theorem intro

Day 2 | March 7 (Fagun 23)

DSA Data Structures & Algorithms	Unit 1 ■ Algorithms & Complexity Introduction to algorithms, Time complexity, Space complexity, Big-O notation, Best/Worst/Average case
NM Numerical Methods	Unit 1 ■ Taylor's Theorem & Error Propagation Review of Taylor series, Propagation of errors through functions, Condition number
CA Computer Architecture	Unit 1 ■ Other Binary Codes & Error Detection Gray code, Excess-3, EBCDIC, Parity bit (odd/even), Parity generator & checker
CG Computer Graphics	Unit 1 ■ Graphics Hardware Raster-scan vs Vector displays, Display processor, Frame buffer, Hard copy devices, Input devices
STAT Statistics II	Unit 1 ■ Estimation Methods of estimation (MLE, MOM), Properties of estimators (Unbiased, Consistent, Efficient), Sample size determination

Day 3 | March 8 (Fagun 24)

DSA Data Structures & Algorithms	Unit 1 ■ Asymptotic Notations Big-O, Big-Omega, Big-Theta notations, Common functions ($\log n$, n , $n \log n$, n^2 , 2^n), Practical examples
NM Numerical Methods	Unit 1 ■ Solving Nonlinear Equations – Bisection Half-interval (Bisection) method, Algorithm steps, Convergence criterion, Error bound formula, Worked example
CA Computer Architecture	Unit 2 ■ Register Transfer Language RTL notation, Register transfer operations, Bus & memory transfers, Control functions
CG Computer Graphics	Unit 1 ■ Graphics Software Software standards (GKS, PHIGS), Machine-independent graphics language, OpenGL overview
STAT Statistics II	Unit 2 ■ Hypothesis Testing Basics Null vs Alternative hypothesis, Type I & Type II errors, Significance level (α), p-value concept, Steps of testing

Day 4 | March 9 (Fagun 25)

DSA Data Structures & Algorithms	Unit 5 ■ Arrays & Array Operations 1D & 2D arrays, Memory layout, Traversal, Insertion & Deletion, Array as ADT
NM Numerical Methods	Unit 1 ■ False Position & Secant Method False position (Regula Falsi) algorithm, Secant method derivation, Comparison with bisection, Convergence rates
CA Computer Architecture	Unit 2 ■ Arithmetic Microoperations Binary adder, Adder-subtractor, Binary incrementer, Arithmetic circuit design
CG Computer Graphics	Unit 2 ■ Scan Conversion – DDA Line DDA (Digital Differential Analyzer) algorithm, Step calculation, Slope cases ($ m < 1$, $ m > 1$), Advantages & limitations
STAT Statistics II	Unit 2 ■ One-Sample Tests Z-test for mean (known variance), t-test for mean (unknown variance), Test for single proportion, Decision rule

Day 5 | March 10 (Fagun 26)

DSA Data Structures & Algorithms	Unit 5 ■ Linked List Introduction Concept of linked list vs array, Node structure (data + pointer), Singly linked list, Head pointer
NM Numerical Methods	Unit 1 ■ Newton-Raphson Method Derivation from Taylor series, Algorithm, Convergence (quadratic), Pitfalls (zero derivative), Multiple roots
CA Computer Architecture	Unit 2 ■ Logic & Shift Microoperations AND/OR/XOR/NOT operations, Hardware implementation, Logical/Circular/Arithmetic shifts, Shifter circuit
CG Computer Graphics	Unit 2 ■ Bresenham's Line Algorithm Decision parameter derivation, Algorithm steps, Integer arithmetic advantage, Comparison with DDA
STAT Statistics II	Unit 2 ■ Two-Sample Tests Test for difference between two means, Two-proportion test, Paired sample t-test, Confidence interval linkage

Day 6 | March 11 (Fagun 27)

DSA Data Structures & Algorithms	Unit 5 ■ Singly Linked List Operations Node creation, Insertion at beginning/end/position, Deletion from beginning/end/position, Traversal, Search
NM Numerical Methods	Unit 1 ■ Fixed Point Iteration & Horner's Method Fixed-point iteration, Convergence condition $ g'(x) < 1$, Horner's synthetic division, Worked examples
CA Computer Architecture	Unit 3 ■ Basic Computer Organization Instruction code, Operation code, Stored program concept, Registers of basic computer, Common bus system
CG Computer Graphics	Unit 2 ■ Midpoint Circle Algorithm Circle equation, Decision parameter derivation, 8-way symmetry, Algorithm steps, Worked example
STAT Statistics II	Unit 3 ■ Non-Parametric Tests – Part 1 Parametric vs non-parametric, When to use non-parametric, Run test, Binomial test, Kolmogorov-Smirnov (one-sample)

Day 7 | March 12 (Fagun 28) ■ REVISION DAY

DSA Data Structures & Algorithms	Review ■ Week 1 Revision – DSA Re-read: ADT, Big-O notation, Array operations, Singly linked list; Solve: 5 past questions on arrays & complexity
NM Numerical Methods	Review ■ Week 1 Revision – Numerical Re-read: All root-finding methods; Solve: 3 numerical problems each for Bisection, Newton-Raphson, False Position
CA Computer Architecture	Review ■ Week 1 Revision – Architecture Re-read: Data representation, RTL, Microoperations, Basic computer organization; Draw block diagrams
CG Computer Graphics	Review ■ Week 1 Revision – Graphics Re-read: DDA, Bresenham's line, Midpoint circle; Re-derive decision parameters from scratch
STAT Statistics II	Review ■ Week 1 Revision – Statistics Re-read: CLT, Hypothesis testing steps, Z-test, t-test; Solve 3 hypothesis testing problems

Week 2 – Core Concepts (Days 8–14)

Day 8 | March 13 (Fagun 29)

DSA Data Structures & Algorithms	Unit 2 ■ Stack – Concept & Operations Stack as ADT, LIFO principle, push/pop/peek operations, Stack using array, Stack using linked list, Overflow & underflow
NM Numerical Methods	Unit 2 ■ Interpolation Introduction & Lagrange Interpolation vs Extrapolation, Lagrange interpolation formula, Uniqueness of interpolating polynomial, Worked examples
CA Computer Architecture	Unit 3 ■ Instruction Format & Instruction Set Instruction format, Addressing modes in basic computer, Instruction set completeness, Control unit, Timing signals
CG Computer Graphics	Unit 2 ■ Midpoint Ellipse Algorithm Ellipse equation, Two-region approach, Decision parameters for each region, 4-way symmetry, Algorithm
STAT Statistics II	Unit 3 ■ Non-Parametric Tests – Part 2 Two-independent sample: Median test, K-S two-sample, Wilcoxon Mann-Whitney test, Chi-square test

Day 9 | March 14 (Fagun 30)

DSA Data Structures & Algorithms	Unit 2 ■ Stack Applications Infix/Postfix/Prefix conversion algorithm, Evaluation of postfix expression, Balancing parentheses, Function call stack
NM Numerical Methods	Unit 2 ■ Newton's Forward & Backward Interpolation Forward difference table, Newton's forward formula, Backward difference table, Newton's backward formula, When to use each
CA Computer Architecture	Unit 3 ■ Instruction Cycle & Memory Reference Fetch-decode-execute cycle, Memory reference instructions, I/O instructions, Program interrupt, Interrupt cycle flowchart
CG Computer Graphics	Unit 2 ■ Area Filling Algorithms Scan-line polygon fill, Inside-outside test (odd-even rule), Boundary-fill algorithm (4-connected, 8-connected), Flood-fill
STAT Statistics II	Unit 3 ■ Paired & Multiple Comparison Tests Wilcoxon signed rank test, Cochran's Q test, Friedman two-way ANOVA, Kruskal-Wallis test, When to use each

Day 10 | March 15 (Chaitra 1)

DSA Data Structures & Algorithms	Unit 3 ■ Queue – Concept & Linear Queue Queue as ADT, FIFO principle, enqueue/dequeue operations, Linear queue using array, Limitations of linear queue
NM Numerical Methods	Unit 2 ■ Newton's Divided Differences & Cubic Spline Divided difference table, Newton's interpolation formula using divided differences, Cubic spline concept, Natural boundary conditions
CA Computer Architecture	Unit 4 ■ Microprogrammed Control Control word, Microprogram, Control memory, Control address register, Sequencer, Address sequencing
CG Computer Graphics	Unit 3 ■ 2D Transformations – Translation & Rotation Translation matrix, Rotation matrix (about origin), Homogeneous coordinates, Composite transformation concept
STAT Statistics II	Unit 4 ■ Multiple Correlation Multiple correlation coefficient R, Partial correlation, Interpretation, Difference from simple correlation

Day 11 | March 16 (Chaitra 2)

DSA Data Structures & Algorithms	Unit 3 ■ Circular Queue & Priority Queue Problem with linear queue, Circular queue concept, front/rear management, Priority queue (ascending & descending), Applications
NM Numerical Methods	Unit 2 ■ Regression – Introduction & Linear Regression vs Interpolation, Least squares method derivation, Simple linear regression ($y=a+bx$), Normal equations, Residuals
CA Computer Architecture	Unit 4 ■ Microinstruction Design Conditional branch in microprogram, Mapping of instructions, Subroutines in microprogram, Microinstruction format, Symbolic microinstructions
CG Computer Graphics	Unit 3 ■ 2D Transformations – Scaling, Reflection, Shearing Scaling matrix (uniform & non-uniform), Reflection about axes/origin, Shearing in X and Y directions, Matrix representation
STAT Statistics II	Unit 4 ■ Multiple Linear Regression Multiple linear regression model, Matrix form, Hypothesis testing of regression (F-test), t-test for individual coefficients, Model adequacy

Day 12 | March 17 (Chaitra 3)

DSA Data Structures & Algorithms	Unit 4 ■ Recursion Principle of recursion, Base case & recursive case, Recursion vs Iteration, Factorial, Fibonacci, GCD, Tower of Hanoi (algorithm + trace)
NM Numerical Methods	Unit 2 ■ Non-Linear Regression Exponential regression ($y=ae^{bx}$), Power regression ($y=ax^b$), Linearization technique, Polynomial regression
CA Computer Architecture	Unit 5 ■ CPU Organization Major components of CPU, CPU organization types (single-bus, multiple-bus), Instruction formats (variable/fixed length), Data transfer
CG Computer Graphics	Unit 3 ■ Composite Transformations & Viewing Homogeneous composite transformations, Window-to-viewport transformation, Viewing pipeline, Coordinate systems
STAT Statistics II	Unit 5 ■ Design of Experiments – CRD Experimental design principles (Replication, Randomization, Local control), CRD: layout, ANOVA table, F-test, Post-hoc

Day 13 | March 18 (Chaitra 4)

DSA Data Structures & Algorithms	Unit 4 ■ Recursion Applications & Tail Recursion Tail recursion optimization, Efficiency of recursion, Memory stack during recursion, Converting recursion to iteration
NM Numerical Methods	Unit 3 ■ Numerical Differentiation Two-point formula, Three-point formula, Differentiating tabulated functions using Newton's differences, Maxima/minima of tabulated functions
CA Computer Architecture	Unit 5 ■ Addressing Modes & RISC vs CISC Immediate, Direct, Indirect, Register, Register Indirect, Relative, Indexed modes; RISC features, CISC features, Pros & cons
CG Computer Graphics	Unit 3 ■ Clipping Algorithms Point clipping, Cohen-Sutherland line clipping (region codes, algorithm), Liang-Barsky line clipping, Sutherland-Hodgman polygon clipping
STAT Statistics II	Unit 5 ■ RBD (Randomized Block Design) RBD layout, ANOVA table, F-test, Efficiency of RBD vs CRD, Missing value estimation (one observation), Advantages & limitations

Day 14 | March 19 (Chaitra 5) ■ REVISION DAY

DSA Data Structures & Algorithms	Review ■ Week 2 Revision – DSA Re-read: Stack (push/pop/applications), Queue (linear, circular, priority), Recursion (TOH); Solve past exam questions
NM Numerical Methods	Review ■ Week 2 Revision – Numerical Re-read: Lagrange, Newton's interpolation, Linear & non-linear regression; Solve 3 interpolation problems
CA Computer Architecture	Review ■ Week 2 Revision – Architecture Re-read: Instruction cycle, Microprogrammed control, RISC vs CISC; Draw instruction cycle flowchart from memory
CG Computer Graphics	Review ■ Week 2 Revision – Graphics Re-read: All 2D transformations, clipping algorithms; Practice composite transformation matrix multiplication
STAT Statistics II	Review ■ Week 2 Revision – Statistics Re-read: Non-parametric tests, multiple regression, CRD; Solve one complete ANOVA problem

Week 3 – Important Topics (Days 15–21)

Day 15 | March 20 (Chaitra 6)

DSA Data Structures & Algorithms	Unit 5 ■ Doubly & Circular Linked List Doubly linked list (forward & backward pointers), Operations in DLL, Circular singly linked list, Circular doubly linked list, Applications
NM Numerical Methods	Unit 3 ■ Numerical Integration – Trapezoidal Rule Newton-Cote's quadrature formula, Trapezoidal rule derivation, Single-segment & multi-segment Trapezoidal rule, Error formula
CA Computer Architecture	Unit 6 ■ Pipelining – Concept & Speedup Parallel processing concept, Flynn's classification (SISD/SIMD/MIMD/MISD), Pipeline concept, Speedup equation, Efficiency
CG Computer Graphics	Unit 4 ■ 3D Transformations 3D Translation, Rotation (about X, Y, Z axes), Scaling, Reflection, Shearing matrices, 3D composite transformations
STAT Statistics II	Unit 5 ■ LSD (Latin Square Design) LSD layout (m×m), ANOVA table for LSD, Missing value estimation, Efficiency of LSD vs RBD, Advantages & limitations

Day 16 | March 21 (Chaitra 7)

DSA Data Structures & Algorithms	Unit 5 ■ Stack & Queue as Linked List Implementing stack using linked list (dynamic), Implementing queue using linked list, Advantages over array implementation
NM Numerical Methods	Unit 3 ■ Simpson's Rules Simpson's 1/3 rule (derivation), Multi-segment Simpson's 1/3 rule, Simpson's 3/8 rule, Multi-segment 3/8 rule, Comparison of accuracy
CA Computer Architecture	Unit 6 ■ Instruction Pipeline Three-segment pipeline (FI, EI, WO), Four-segment pipeline, Pipeline conflicts: data hazard, control hazard, structural hazard; Solutions
CG Computer Graphics	Unit 4 ■ 3D Viewing & Projections 3D viewing pipeline, World-to-screen transformation, Orthographic projection, Parallel projection, Perspective projection (vanishing points)
STAT Statistics II	Unit 6 ■ Stochastic Processes – Introduction Definition, Classification of stochastic processes, Markov chain definition, Transition probability matrix, Chapman-Kolmogorov equation

Day 17 | March 22 (Chaitra 8)

DSA Data Structures & Algorithms	Unit 6 ■ Sorting – Bubble & Selection Sort Bubble sort (algorithm, trace, $O(n^2)$), Selection sort (algorithm, trace, $O(n^2)$), Comparison, Stability
NM Numerical Methods	Unit 3 ■ Gaussian & Romberg Integration Gaussian integration algorithm, Gauss-Legendre points and weights, Romberg integration (Richardson extrapolation), Accuracy comparison
CA Computer Architecture	Unit 6 ■ Vector Processing & Computer Arithmetic Vector operations, Matrix multiplication in pipeline, Addition/subtraction with signed magnitude, 2's complement arithmetic
CG Computer Graphics	Unit 5 ■ 3D Object Representation – Polygon Surfaces Polygon tables (vertex, edge, polygon), Surface normals, Plane equations, Spatial orientation, Polygon meshes, Wireframe
STAT Statistics II	Unit 6 ■ Markov Chains & Steady State Matrix approach, n-step transition probabilities, Steady-state distribution calculation, Absorbing states, Worked examples

Day 18 | March 23 (Chaitra 9)

DSA Data Structures & Algorithms	Unit 6 ■ Insertion Sort & Shell Sort Insertion sort (algorithm, trace, $O(n^2)$ worst, $O(n)$ best), Shell sort (gap sequence, algorithm, $O(n^{1.5})$), When to prefer each
NM Numerical Methods	Unit 4 ■ System of Linear Equations – Gaussian Elimination Review of matrix properties, Gaussian elimination algorithm, Partial pivoting, Back substitution, Gauss-Jordan method
CA Computer Architecture	Unit 7 ■ Booth Multiplication & Division Booth's multiplication algorithm (trace with example), Division of signed magnitude data, Divide overflow detection
CG Computer Graphics	Unit 5 ■ Curves – Bezier & B-Spline Parametric cubic curves, Spline representation, Bezier curve (Bernstein basis, control points, properties), B-spline curve
STAT Statistics II	Unit 6 ■ Counting & Queuing Processes Binomial process, Poisson process properties, Queuing system components, Little's law, Bernoulli single-server queue

Day 19 | March 24 (Chaitra 10)

DSA Data Structures & Algorithms	Unit 6 ■ Divide & Conquer – Merge Sort & Quick Sort Merge sort (divide-merge, $O(n \log n)$), Quick sort (partition, pivot selection, $O(n \log n)$ avg), Comparison
NM Numerical Methods	Unit 4 ■ Matrix Factorization & Inverse LU decomposition (Doolittle algorithm), Cholesky's algorithm (symmetric positive definite), Matrix inverse using Gauss-Jordan
CA Computer Architecture	Unit 8 ■ Input-Output Organization I/O interface, I/O vs memory bus, Isolated vs memory-mapped I/O, Asynchronous transfer (strobe, handshaking)
CG Computer Graphics	Unit 6 ■ Solid Modeling Sweep representation, Boundary representation (B-rep), Space partitioning, BSP trees (Binary Space Partition), Octree representation
STAT Statistics II	Unit 6 ■ M/M/1 Queue & Simulation M/M/1 queuing system, Performance measures (L_q , W_q , L , W), Simulation of stochastic processes, Worked example

Day 20 | March 25 (Chaitra 11)

DSA Data Structures & Algorithms	Unit 6 ■ Heap Sort & Sorting Efficiency Heap data structure, Max-heap & min-heap, Heapify, Heap sort algorithm ($O(n \log n)$), Efficiency comparison of all sorting algorithms
NM Numerical Methods	Unit 4 ■ Iterative Methods – Jacobi & Gauss-Seidel Jacobi iteration (algorithm, convergence condition), Gauss-Seidel method (faster convergence), Diagonally dominant systems
CA Computer Architecture	Unit 8 ■ Modes of Data Transfer & DMA Programmed I/O, Interrupt-initiated I/O, DMA (concept, DMA controller, cycle stealing), Priority interrupt (polling, daisy-chain, parallel)
CG Computer Graphics	Unit 7 ■ Visible Surface Detection Image space vs object space techniques, Back-face detection, Depth buffer (Z-buffer) algorithm, A-buffer, Scan-line algorithm
STAT Statistics II	Unit 1 & 2 ■ Formula Revision – Stats CLT formula, Confidence interval formulas, Z-test, t-test, chi-square test statistics; Write all on one page

Day 21 | March 26 (Chaitra 12) ■ REVISION DAY

DSA Data Structures & Algorithms	Review ■ Week 3 Revision – DSA Re-read: All sorting algorithms; Trace bubble sort, merge sort, quick sort on sample array; Write time complexities
NM Numerical Methods	Review ■ Week 3 Revision – Numerical Re-read: Integration rules, Gaussian elim, Jacobi/Gauss-Seidel; Solve one full linear system problem
CA Computer Architecture	Review ■ Week 3 Revision – Architecture Re-read: Pipeline, I/O organization, DMA; Draw pipeline diagram with 4 stages
CG Computer Graphics	Review ■ Week 3 Revision – Graphics Re-read: 3D transformations, projections, BSP, Z-buffer; Practice 3D rotation matrix problems
STAT Statistics II	Review ■ Week 3 Revision – Statistics Re-read: LSD, Stochastic process, M/M/1; Solve one complete experimental design ANOVA

Week 4 – Advanced Topics (Days 22–25)

Day 22 | March 27 (Chaitra 13)

DSA Data Structures & Algorithms	Unit 7 ■ Searching – Sequential & Binary Search Sequential (linear) search $O(n)$, Binary search (iterative & recursive) $O(\log n)$, Efficiency comparison, Sentinel search
NM Numerical Methods	Unit 4 ■ Eigenvalues & Power Method Eigenvalue & eigenvector definition, Characteristic equation, Power method algorithm, Dominant eigenvalue, Worked example
CA Computer Architecture	Unit 9 ■ Memory Organization – Hierarchy & Cache Memory hierarchy (register, cache, RAM, disk), Cache memory, Locality of reference, Hit & miss ratio, Mapping techniques (direct, associative, set-associative)
CG Computer Graphics	Unit 7 ■ Depth Sorting & Ray Tracing Painter's algorithm (depth sorting), BSP tree method for visibility, Octree visibility, Ray tracing concept, Shadow rays
STAT Statistics II	Review ■ Comprehensive Formula Sheet Write down ALL formulas: test statistics, ANOVA tables, regression equations, Markov chain matrices

Day 23 | March 28 (Chaitra 14)

DSA Data Structures & Algorithms	Unit 7 ■ Hashing Hash function types (division, folding, mid-square), Hash tables, Collision resolution (chaining, linear probing, quadratic probing, double hashing)
NM Numerical Methods	Unit 5 ■ Ordinary Differential Equations – Euler's Methods Review of ODE types, Initial value problem, Taylor series method, Picard's method, Euler's method (algorithm, accuracy), Heun's method
CA Computer Architecture	Unit 9 ■ Main Memory & Associative Memory RAM & ROM chips, Memory address map, Memory-CPU connection, Associative memory (hardware organization, match logic, read/write)
CG Computer Graphics	Unit 8 ■ Illumination Models Ambient light, Diffuse reflection (Lambert's law), Specular reflection, Phong model (complete formula), Intensity attenuation, Transparency
STAT Statistics II	Review ■ Past Questions – Statistics Solve 5 past exam questions covering hypothesis testing, regression, and experimental design

Day 24 | March 29 (Chaitra 15)

DSA Data Structures & Algorithms	Unit 8 ■ Trees – Introduction & Binary Tree Tree terminology (root, leaf, height, depth, level), Binary tree properties, Full & complete binary trees, Basic operations
NM Numerical Methods	Unit 5 ■ Runge-Kutta Methods Classical RK4 method (derivation, algorithm), Comparison of Euler vs RK2 vs RK4 accuracy, System of ODEs
CA Computer Architecture	Unit 9 ■ Cache Write Policies Write-through policy, Write-back (copy-back) policy, Write allocation, Cache coherence basics, Auxiliary memory (disk, tape)
CG Computer Graphics	Unit 8 ■ Surface Rendering Methods Constant-intensity (flat) shading, Gouraud shading (vertex normals, interpolation), Phong shading (normal interpolation), Fast Phong shading
STAT Statistics II	Review ■ Past Questions – All Subjects Solve 2 questions each from: hypothesis testing, non-parametric, regression, DOE, stochastic process

Day 25 | March 30 (Chaitra 16)

DSA Data Structures & Algorithms	Unit 8 ■ BST & AVL Tree Binary Search Tree (insert, delete, search, traversals – inorder/preorder/postorder), AVL tree, Balance factor, Rotations (LL, RR, LR, RL)
NM Numerical Methods	Unit 5 ■ Boundary Value Problems & PDEs Shooting method, Review of PDEs, Classification (elliptic, parabolic, hyperbolic), Finite difference equations, Laplace equation, Poisson equation
CA Computer Architecture	Review ■ Full Architecture Revision Go through ALL units; focus on: pipeline speedup formula, Booth multiplication trace, cache mapping; Write short notes
CG Computer Graphics	Unit 9 & 10 ■ Virtual Reality & OpenGL VR concept, VR system components, Types of VR, Position trackers; OpenGL basics, Callback functions, Color commands, Drawing primitives
STAT Statistics II	Review ■ Non-Parametric & DOE Revision Re-read all non-parametric tests with conditions; Redo CRD, RBD, LSD ANOVA tables from scratch

Final Revision (Days 26–30)

Day 26 | March 31 (Chaitra 17) ■ REVISION DAY

DSA Data Structures & Algorithms	Unit 8 ■ Graph Theory Graph definition, Types (directed/undirected, weighted), Adjacency matrix, Adjacency list, BFS algorithm, DFS algorithm
NM Numerical Methods	Review ■ All Methods Quick Recap One-page summary of ALL numerical methods with formulas; Focus on methods that appear most in past papers
CA Computer Architecture	Review ■ Architecture Short Notes Create cheat sheet: RTL, pipeline stages, cache formulas, Booth algorithm steps, I/O modes
CG Computer Graphics	Review ■ Graphics Algorithm Summary Write all algorithms: DDA, Bresenham, Midpoint circle, Scan-line fill, all 2D & 3D transformation matrices, Clipping
STAT Statistics II	Review ■ Statistics Quick Revision Re-read sampling, all tests; Make formula card: when to use Z/t/F/chi-square test

Day 27 | April 1 (Chaitra 18) ■ REVISION DAY

DSA Data Structures & Algorithms	Unit 8 ■ Graph Algorithms Minimum Spanning Tree: Kruskal's algorithm (greedy, union-find), Prim's algorithm; Shortest Path: Dijkstra's algorithm (trace example)
NM Numerical Methods	Review ■ Numerical Problem Practice Solve past exam: 1 root-finding, 1 interpolation, 1 integration, 1 linear system, 1 ODE problem – time yourself
CA Computer Architecture	Review ■ Architecture Past Questions Solve 5 past board exam questions on architecture; Focus on: data representation, pipeline, Booth multiplication
CG Computer Graphics	Review ■ Graphics Past Questions Solve 5 past questions: line/circle algorithms, 2D transformations, clipping, illumination models
STAT Statistics II	Review ■ Statistics Past Questions Solve 5 past questions: hypothesis tests, ANOVA, regression, non-parametric

Day 28 | April 2 (Chaitra 19) ■ REVISION DAY

DSA Data Structures & Algorithms	Review ■ DSA Full Revision Pass Go through ALL units: ADT → Arrays → Stack → Queue → Recursion → Linked List → Sorting → Searching → Trees → Graphs
NM Numerical Methods	Review ■ Numerical Full Revision Pass Go through ALL units: Errors → Root finding → Interpolation → Regression → Differentiation → Integration → Linear systems → ODE → PDE
CA Computer Architecture	Review ■ Architecture Full Revision Pass Go through ALL units: Data representation → RTL → Basic computer → Microprogrammed → CPU → Pipeline → Arithmetic → I/O → Memory
CG Computer Graphics	Review ■ Graphics Full Revision Pass Go through ALL units: Intro → Scan conversion → 2D transform → 3D transform → 3D objects → Solid modeling → Visibility → Illumination → VR → OpenGL
STAT Statistics II	Review ■ Statistics Full Revision Pass Go through ALL units: Sampling → Hypothesis testing → Non-parametric → Regression → DOE → Stochastic process

Day 29 | April 3 (Chaitra 20) ■ REVISION DAY

DSA Data Structures & Algorithms	Review ■ DSA Mock Exam Practice Solve a full 3-hour mock exam paper covering all DSA topics; Check answers and note weak areas
NM Numerical Methods	Review ■ Numerical Mock Exam Practice Solve a complete numerical methods paper; Focus on accuracy of calculation in integration & interpolation
CA Computer Architecture	Review ■ Architecture Mock Exam Practice Solve a complete architecture paper; Focus on: trace Booth multiplication, pipeline diagrams, cache calculations
CG Computer Graphics	Review ■ Graphics Mock Exam Practice Solve a complete graphics paper; Focus on: algorithm traces (DDA, Bresenham, Midpoint), transformation matrices
STAT Statistics II	Review ■ Statistics Mock Exam Practice Solve a complete statistics paper; Check: correct test selection, ANOVA table completion, p-value interpretation

Day 30 | April 4 (Chaitra 21) ■ REVISION DAY

DSA Data Structures & Algorithms	Review ■ Final Day – DSA Key Points Review only your short notes; Key formulas: Big-O table, BST operations, Graph algorithm complexity
NM Numerical Methods	Review ■ Final Day – Numerical Key Points Review formula sheet only; Key: Newton-Raphson formula, Trapezoidal & Simpson formula, RK4 formula
CA Computer Architecture	Review ■ Final Day – Architecture Key Points Review short notes only; Key: Pipeline speedup = $n/(1+(n-1)/k)$, Cache hit ratio formula, Booth algorithm
CG Computer Graphics	Review ■ Final Day – Graphics Key Points Review transformation matrices only; Key: DDA step formula, Bresenham decision parameter, Cohen-Sutherland codes
STAT Statistics II	Review ■ Final Day – Statistics Key Points Review formula card only; SLEEP EARLY – rest is as important as study tonight!

■ Exam Success Tips

■ Study Method	Use Pomodoro: 50 min study, 10 min break. Write notes by hand — don't just read. Solve problems actively.
■ Numerical Method	Always write the formula first, then substitute. Show all steps. Practice at least 2 problems per method daily.
■ DSA	Draw diagrams for linked lists, trees, graphs. Trace algorithms step by step on paper. Memorize time complexities.
■■ Computer Architecture	Draw block diagrams. Trace Booth multiplication manually. Memorize pipeline speedup formula.
■ Computer Graphics	Derive decision parameters. Practice transformation matrices. Know which algorithm to use for which shape.
■ Statistics	Memorize when to use each test. Write ANOVA tables in practice. Know all non-parametric test conditions.
■ Last 3 Days	Only review notes — no new topics. Solve past papers. Sleep 7-8 hours. Eat well. Stay calm.

You can do this! Best of luck for your Board Exam! ■